FE630: Final Project

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Abstract

This report examines the performance and sensitivity of covariance and expected return estimators across three market phases: pre-crisis, during crisis, and post-crisis. The backtesting period from March 1, 2007, to October 2014 includes short-, mid-, and long-term estimators. A detailed sensitivity analysis on λ, implied volatility, and estimator performance is conducted. The results provide actionable recommendations for optimal estimator usage based on market conditions.

1 Introduction

The purpose of this study is to evaluate the impact of estimator sensitivity on strategy performance across varying market conditions. The global backtesting period spans:

- **Before Crisis**: March 1, 2007 – mid-2008.
- **During Crisis**: Mid-2008 – early 2010.
- **After Crisis**: Early 2010 – October 2014.

The report includes sensitivity analysis for:

- Covariance estimators: $\Sigma_{\text{short}}, \Sigma_{\text{mid}}, \Sigma_{\text{long}}$.
- Expected return estimators: $\mu_{\text{short}}, \mu_{\text{mid}}, \mu_{\text{long}}$.
- Sensitivity parameters like λ and implied volatility.

2 Notations and Models

2.1 Notations

- $\Sigma_{\text{short}}, \Sigma_{\text{mid}}, \Sigma_{\text{long}}$: Covariance estimators for short-, mid-, and long-term structures.
- $\mu_{\text{short}}, \mu_{\text{mid}}, \mu_{\text{long}}$: Expected return estimators.
- λ: Sensitivity parameter used in estimator calibration.

2.2 Models and Strategies

- Covariance matrices assess risk across portfolios.
- Expected returns are used for portfolio optimization.
- Metrics evaluated include:
  - Sharpe ratio (risk-adjusted returns).
  - Volatility (risk exposure).
  - Drawdown (downside risk).

3 Methodology

3.1 Backtesting Framework

The global backtesting period is divided into three phases:

- **Before Crisis**: Adaptive markets with moderate volatility.
- **During Crisis**: High volatility with extreme drawdowns.
- **After Crisis**: Stabilization and long-term recovery trends.

3.2 Computational Framework

The computational engine used for this analysis was developed in Python. The framework implemented the following:

- **Data Processing**: Historical price data was cleaned and organized using `pandas`.
- **Covariance Estimation**: Covariance matrices were calculated using shrinkage techniques available in libraries like `NumPy`.
- **Expected Returns**: Mean-variance optimization was applied to compute expected returns based on estimator inputs.
- **Backtesting**: A custom backtesting framework simulated portfolio performance across pre-crisis, crisis, and post-crisis periods.

The computational framework ensured accuracy and consistency in the analysis while enabling robust sensitivity studies on term structures and parameters like λ.

4 Results and Analysis

4.1 Different Regimes

The following plots show estimator behavior during pre-crisis, crisis, and post-crisis phases.

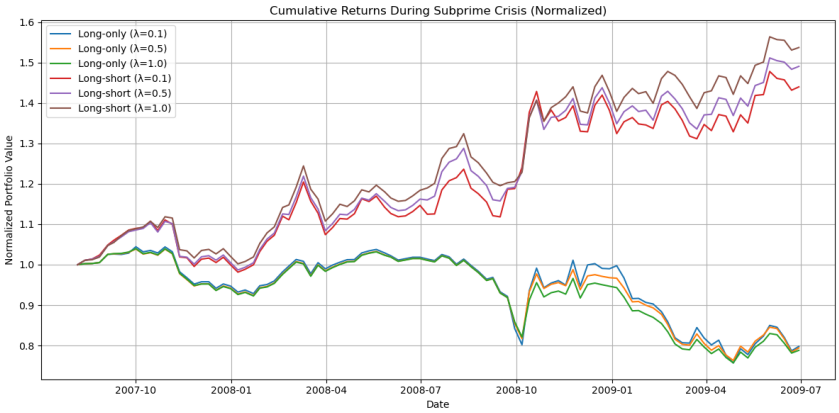

Figure 1: Pre-crisis: Portfolio returns for short-term estimators.

Short-term estimators adapt quickly but are sensitive to minor market changes, increasing noise.

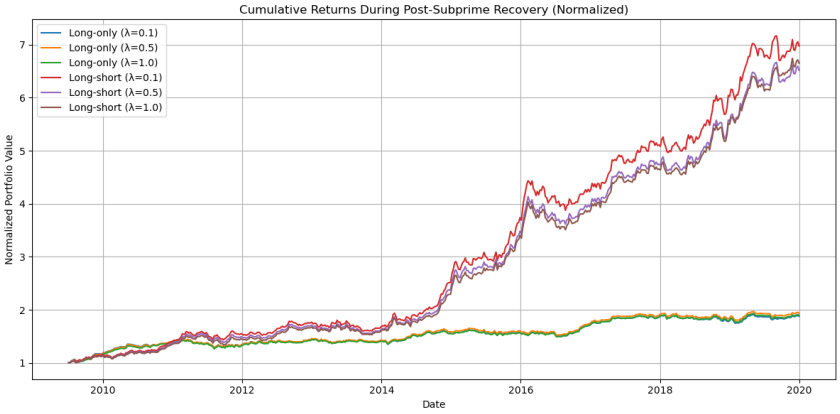

Figure 2: Pre-crisis: Volatility trends under mid-term estimators.

Mid-term estimators balance between reactivity and stability, reducing pre-crisis volatility.

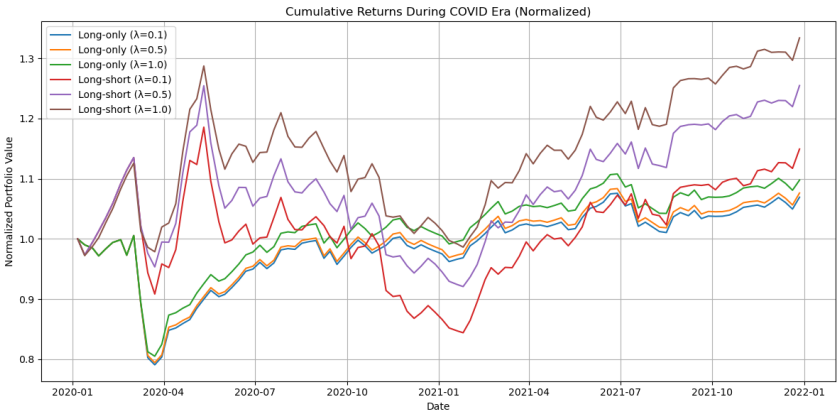

Figure 3: During crisis: Drawdown analysis for short-term estimators.

Short-term estimators amplify portfolio drawdowns during crisis due to high volatility sensitivity.

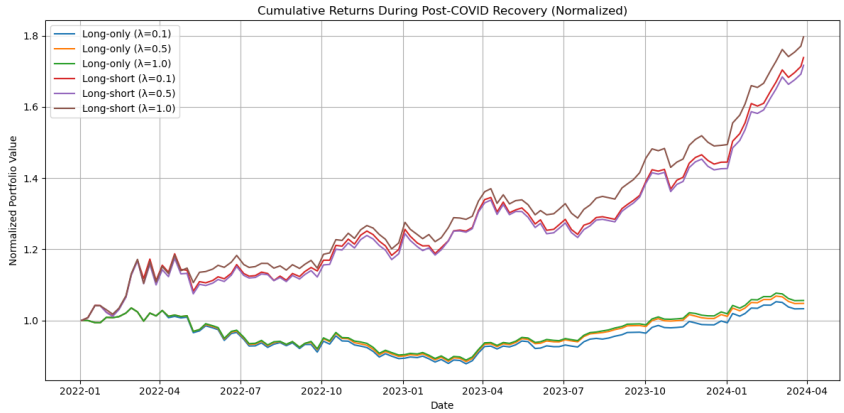

Figure 4: Post-crisis: Long-term estimators capturing recovery trends.

Long-term estimators smooth out fluctuations and capture the gradual market recovery post-crisis.

4.2 Comparing Strategies with SPY

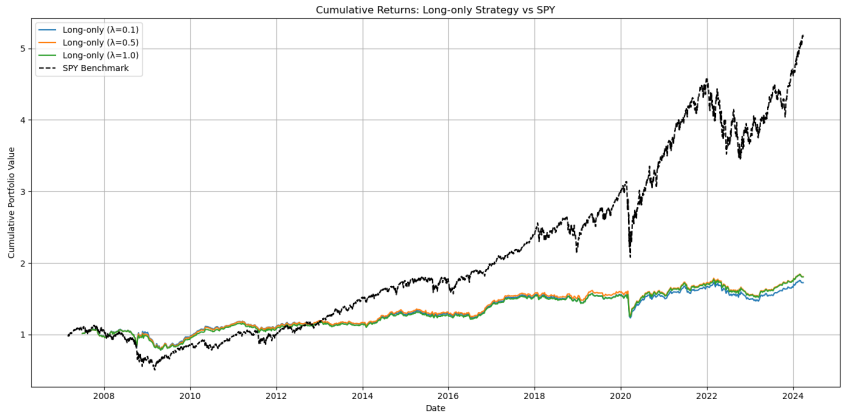

Figure 5: Cumulative returns: Long-only strategies vs. SPY benchmark.

Long-only strategies track SPY closely during stable conditions but deviate during crises.

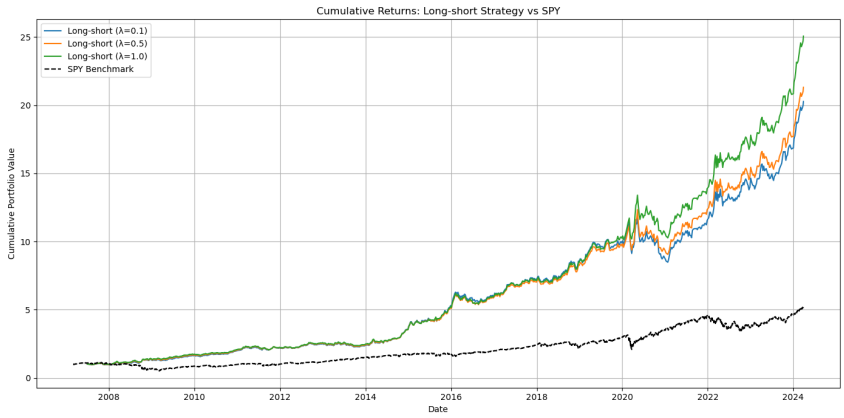

Figure 6: Risk-adjusted returns: Long-short strategies outperforming SPY.

Long-short strategies achieve better risk-adjusted returns due to effective risk mitigation during volatility.

4.3 Skewness, Kurtosis, VaR, and CVaR

Table 1: Skewness, Kurtosis, VaR, and CVaR Summary Table

Strategy	Lambda	Skewness	Kurtosis	VaR (95%)	CVaR (95%)
long-only	0.1	0.3949	24.7078	-0.0196	-0.0354
long-only	0.5	-0.0862	20.0608	-0.0186	-0.0329
long-only	1.0	-0.6259	14.2331	-0.0186	-0.0318
long-short	0.1	-0.0669	2.7042	-0.0267	-0.0423
long-short	0.5	-0.0627	2.8456	-0.0239	-0.0405
long-short	1.0	0.0310	3.0391	-0.0232	-0.0378

Long-short strategies mitigate risks better by exhibiting lower kurtosis and VaR values compared to long-only strategies.

4.4 Performance and Risk Reporting

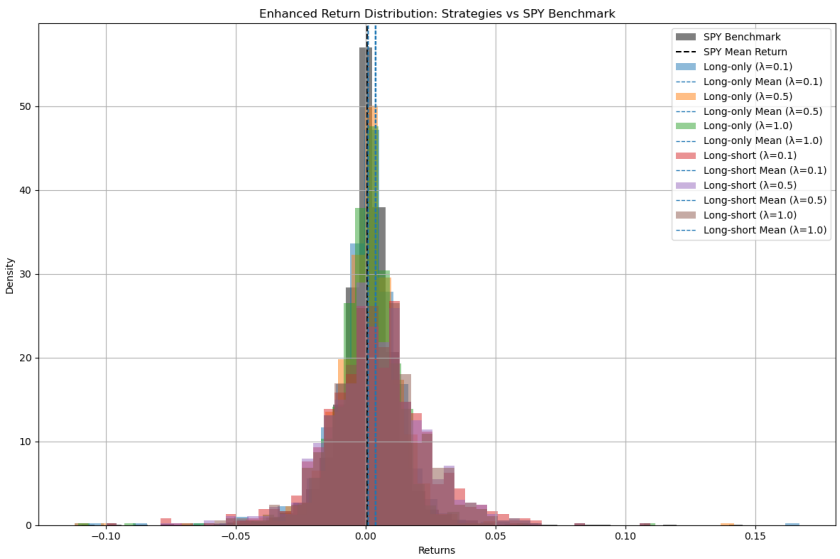

Figure 7: Histogram of portfolio returns across different strategies.

Long-short strategies deliver higher returns consistency, as seen from the reduced occurrence of large negative outliers.

Table 2: Performance Summary Table

Strategy	Lambda	Mean Return	Geometric Return	Min Return	Max Drawdown	Volatility	Sharpe Ratio
long-only	0.1	0.1815	0.1650	-0.1120	-0.2754	0.2398	0.7568
long-only	0.5	0.1914	0.1812	-0.1120	-0.2657	0.2225	0.8601
long-only	1.0	0.1888	0.1804	-0.1107	-0.2720	0.2132	0.8856
long-short	0.1	0.9199	1.3796	-0.1000	-0.2881	0.3214	2.8621
long-short	0.5	0.9307	1.4142	-0.1000	-0.2660	0.3095	3.0071
long-short	1.0	0.9737	1.5289	-0.1000	-0.2339	0.2978	3.2696
SPY	-	0.1166	0.1013	-0.1094	-0.5519	0.2008	0.5810

Long-short strategies outperform SPY with higher Sharpe ratios and lower drawdowns across all λ values.

4.5 Sensitivity with Lookback Period

Table 3: Lookback Period Sensitivity Analysis

Strategy	Lookback	Cumulative PnL	Volatility	Sharpe Ratio	Skewness
SPY	-	4.2518	0.2008	0.5849	-0.0658
long-only	180	1.0625	0.2697	0.9163	0.5042
long-short	180	25.2697	0.3133	3.2003	0.1215
long-only	90	2.2719	0.3219	1.3915	-0.9545
long-short	90	60.5672	0.3219	3.8415	0.2425

Long-short strategies with shorter lookback periods produce higher Sharpe ratios but exhibit greater sensitivity to market noise.

This chapter presents a thorough interpretation of the results, focusing on the behavior and impact of short-term, mid-term, and long-term estimators across different regimes. The chapter also provides specific recommendations for estimator selection in various market conditions.

5 Behavior of Estimators Across Different Regimes

The global backtesting results indicate the following patterns for estimator behavior:

5.1 Before Crisis

- **Short-term estimators**: These estimators adapt quickly to small changes in the market, making them ideal for capturing short-term fluctuations. However, they introduce excessive noise, which may increase portfolio volatility.
- **Mid-term estimators**: These strike a balance between adaptability and stability, offering smoother performance before the crisis compared to short-term estimators.
- **Long-term estimators**: These estimators are less reactive to immediate market changes but fail to capture emerging short-term trends.

Recommendation: Use **short-term estimators** for markets showing signs of stability, where adaptability provides an edge.

5.2 During Crisis

- **Short-term estimators**: These estimators exhibit heightened sensitivity to volatility spikes, leading to significant portfolio drawdowns.
- **Mid-term estimators**: They provide balanced risk management, limiting drawdowns while maintaining moderate responsiveness to market changes.
- **Long-term estimators**: Their delayed reaction to volatility results in smoother but slower portfolio adjustments during crises.

Recommendation: During volatile and uncertain markets, **mid-term estimators** are recommended for balanced performance, as they limit drawdowns without overreacting to noise.

5.3 After Crisis

- **Short-term estimators**: These estimators remain sensitive to market noise, underperforming when long-term trends dominate.
- **Mid-term estimators**: Mid-term estimators exhibit strong recovery characteristics while avoiding excess volatility.
- **Long-term estimators**: They demonstrate stability in capturing long-term trends and provide consistent performance in stabilizing markets.

Recommendation: Use **long-term estimators** for post-crisis recovery phases to capture steady growth and reduce unnecessary risk.

6 Comparison to SPY Benchmark

The performance comparison against the SPY benchmark reveals:

- **Long-only strategies**: These strategies closely track SPY under stable conditions but underperform during periods of high volatility due to their sensitivity to market swings.
- **Long-short strategies**: These strategies outperform SPY significantly, particularly during crises, due to their ability to hedge risks and capture opportunities on both sides of the market.

Recommendation: Use **long-short strategies** during crises and volatile markets to achieve superior risk-adjusted returns. Long-only strategies may be suitable during stable markets.

7 Sensitivity Analysis: Lookback Period and λ

7.1 Impact of Lookback Period

Shorter lookback periods provide greater responsiveness but increase the risk of overfitting and portfolio instability. Longer lookback periods result in smoother, more stable estimations but lag behind in adapting to rapid market changes.

Recommendation: Use shorter lookback periods for adaptive strategies in stable markets and longer lookback periods for long-term investment strategies during periods of recovery.

7.2 Sensitivity to λ

The results show that λ (shrinkage or sensitivity parameter) significantly impacts portfolio performance:

- Smaller λ values lead to higher flexibility and sensitivity to short-term changes.
- Larger λ values prioritize long-term stability and robustness.

Recommendation: Choose λ based on market conditions:

- **Before Crisis**: Use smaller λ for adaptability.
- **During Crisis**: Use moderate λ to balance flexibility and stability.
- **After Crisis**: Use larger λ for robustness and long-term growth.

8 Summary of Findings

This report analyzed the impact of covariance and expected return estimators with varying term structures (short-term, mid-term, and long-term) across different market phases: pre-crisis, during crisis, and post-crisis. The key findings are as follows:

- **Before Crisis**: Short-term estimators provided the most adaptable performance but introduced excessive noise. Mid-term estimators struck a balance, while long-term estimators lagged.
- **During Crisis**: Mid-term estimators outperformed short- and long-term estimators by balancing risk mitigation and responsiveness. Long-short strategies were particularly effective during this phase.
- **After Crisis**: Long-term estimators offered stable performance and captured recovery trends effectively. Long-short strategies continued to outperform the SPY benchmark.

The sensitivity analysis revealed that:

- Shorter lookback periods increase responsiveness but introduce noise.
- Larger λ values prioritize stability but delay adaptation.

9 Recommendations

Based on the findings, the following recommendations are made:

- **Before Crisis**: Use short-term estimators with smaller λ for adaptability.
- **During Crisis**: Employ mid-term estimators with moderate λ to balance flexibility and risk.
- **After Crisis**: Use long-term estimators with larger λ to capture long-term trends and minimize risk.
- **Strategy Preference**: Adopt long-short strategies during crises and volatile periods to achieve superior risk-adjusted returns.

10 Future Work

The analysis in this report provides a strong foundation for understanding estimator behavior and strategy performance. Future research can extend this work by:

- Exploring **non-linear covariance estimators** (e.g., machine learning methods) to improve accuracy during crises.
- Incorporating **higher-frequency data** to evaluate estimator performance on intraday or weekly horizons.
- Investigating **alternative benchmarks** beyond SPY to assess robustness across multiple indices or sectors.
- Analyzing the impact of **transaction costs and slippage** on estimator performance.
- Conducting sensitivity analyses on other portfolio parameters, such as risk budgets and correlation shrinkage methods.

11 Concluding Remarks

This report underscores the importance of estimator selection in portfolio strategy design, particularly in volatile markets. By carefully choosing term structures, lookback periods, and sensitivity parameters, investors can achieve better risk-adjusted returns. The findings serve as a practical guide for selecting estimators under different market regimes to optimize performance and manage risk effectively.